

Boolean logic

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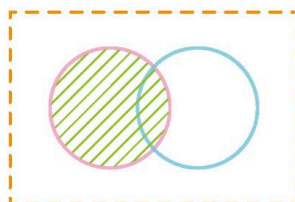
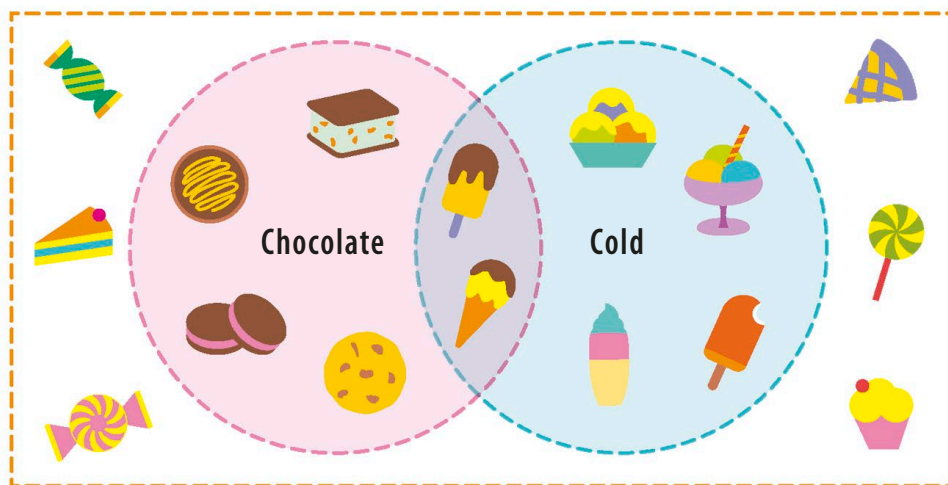
Boolean logic, also called Boolean algebra, is a branch of mathematics with only two values – 0 and 1. Created by English mathematician George Boole, it plays a critical role in circuit design and application design.

Boolean operators

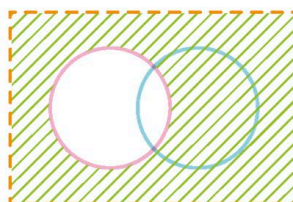
Boolean logic is all about collections of objects, called sets. Each set can be further separated into subsets. For example, the set of all desserts can be separated into subsets like chocolate desserts and cold desserts. Some desserts, such as chocolate ice cream, belong to both groups. Others, such as ginger cake, belong to neither. While regular algebra has operators, such as add (+), subtract (–), and multiply (*), Boolean algebra has AND, OR, and NOT. These operators can be used to create more specific subgroups.

▷ Dessert Venn diagram

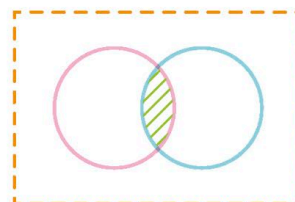
This Venn diagram shows how a set of different desserts can be divided into subsets of chocolate desserts and cold desserts. If a dessert belongs to a group, it corresponds to the Boolean value 1. Otherwise, it's 0.



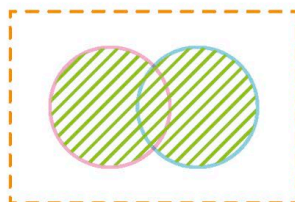
Chocolate



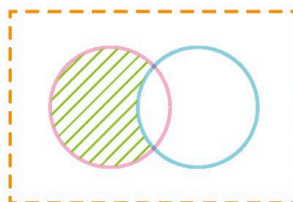
NOT chocolate



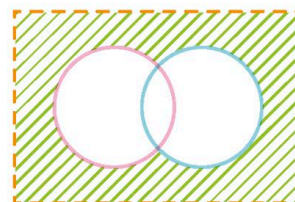
Chocolate AND cold



Chocolate OR cold



Chocolate AND (NOT cold)



(NOT cold) AND (NOT chocolate)

▷ Other possibilities

A specific type of dessert can be isolated using the right combination of Boolean operators.